A Storm Surge Prediction Model for the Northern Bay of Bengal with Application to the Cyclone Disaster in April 1991

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ABSTRACT

A numerical model for simulating and predicting tides and storm surges in regions that include areas of open sea combined with estuarine channels and intertidal banks is described. The model makes use of modified depth-averaged equations with a numerical scheme in which the solution of 1D equations for narrow channels, 2D equations for the open sea, and an approach to modeling inundation are combined within a unified framework. This simplifies the construction of realistic models for complex coastal areas and deltas.

The model is applied to the northern shelf of the Bay of Bengal and the Ganges Delta, including the coast of Bangladesh devastated by a cyclone surge in April 1991 in which about 140 000 people died. Hindcast and "forecast" simulations of this event, using forcing derived from a semianalytical cyclone model with data supplied by the Joint Typhoon Warning Center (JTWC), are described. It is shown that the timing of cyclone landfall and its coincidence with high tide determine the area worst affected by flooding. The "forecast" was based on a standard JTWC warning issued 12–18 hours ahead of the event. The predicted landfall was located accurately but its timing was about 5 hours late. As a result, the area of highest water levels shifted north. Improvements in the model and procedures required to produce useful warnings are discussed.

Finally, results for the April 1991 storm are compared with those from a simulation of the great cyclone in November 1970. Peak surges and water levels were similar in magnitude in the two events, but differences in track and tidal conditions are again shown to be important, producing flooding in a large area of the southern delta in 1970 but along the mainland coast south of Chittagong in 1991.

1. Introduction

Coastal floods produced by cyclone-generated storm surges are a major cause of loss of life in the coastal areas surrounding the Bay of Bengal [see Murty et al. (1986) for a review of the problem]. For a number of reasons Bangladesh is particularly vulnerable. First, it lies in the path of tropical cyclones, which typically originate in the central and southern parts of the bay or in the Andaman Sea. The cyclones often follow tracks toward the northwest or north before finally curving east into the northeast corner of the bay and across the coast of Bangladesh. Second, it has a wide and shallow continental shelf (see Fig. 1); strong winds acting over shallow water are the primary generating mechanism for the storm surges, which in conjunction with the substantial astronomical tides can produce exceptionally high water levels. Third, the coastal areas of Bangladesh comprise low-lying and poorly protected land, much of it being the delta of the Ganges and Brahmaputra rivers. Coastal defenses, typically taking the form of earth banks, are designed primarily to prevent flooding by the normal tides and are overtopped or seriously damaged by a cyclone surge. Fourth, the coastal regions support a large population, many of them the poorest people forced to live in vulnerable areas in order to feed themselves by farming and fishing.

All the ingredients for a major disaster are present, and such disasters have occurred on a number of occasions, notably in November 1970 when approximately 250 000 people were killed. In April 1991 a comparable storm struck Bangladesh, with fatalities estimated at 140 000.

Mitigation of these disasters requires a number of complementary and coordinated actions. Improved forecasts of water levels, better warnings, and provision of refuges or shelters in appropriate locations could significantly reduce casualties in future cyclone floods.

Storm surge models have been developed for many parts of the world, and have long been used to provide routine flood warnings in several countries (see, e.g., Flather et al. 1991). A number of tide and surge models already exist for the Bay of Bengal, and have been used to investigate various aspects of its response to cyclones. Notable among them is the pioneering work of Das et al. (1974), who established a model solving the linearized equations and from simulations of a variety of storms derived results that were incorporated into a simple scheme to estimate the maximum surge height

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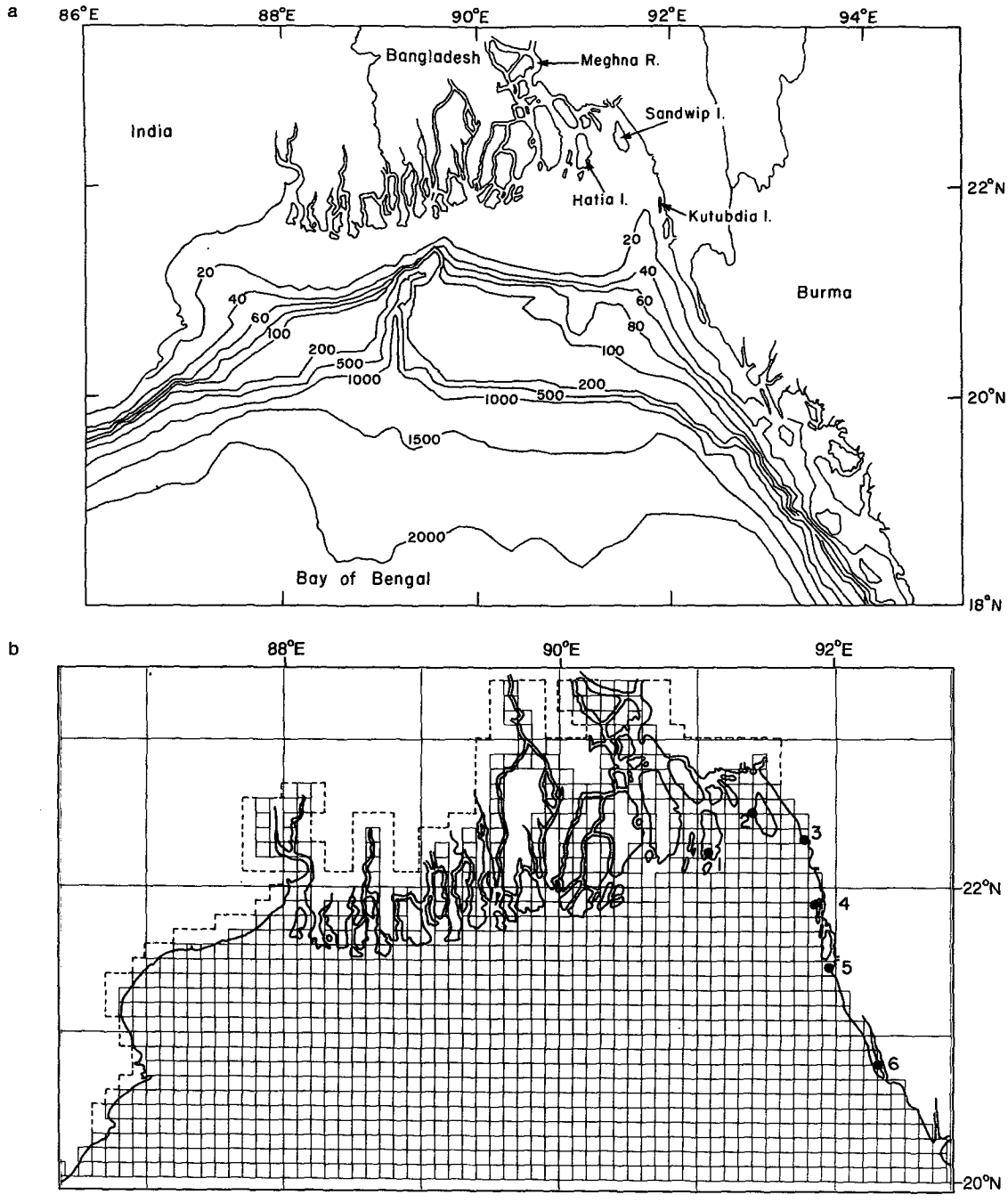

FIG. 1. (a) Map of the northern Bay of Bengal with contoured water depths (m) based on the bathymetry used in the numerical model. (b) Finite-difference grid for the model of the northern Bay of Bengal, with locations of tide gauges referred to in the text (1 = Char Changa; 2 = Sandwip Island; 3 = Chittagong; 4 = Kutubdia Island; 5 = Cox's Bazar; 6 = Shapuri Island). The broken line indicates the limit of the computational area within which flooding can occur.

from basic cyclone parameters. Johns and coworkers (e.g., see Johns et al. 1981; Johns et al. 1985) developed a series of models distinguished by their use of a transformation of one horizontal space coordinate, allowing a stretching and hence refinement of the computational grid near shore. The models were used to clarify the role of bathymetry, inundation, and the presence of rivers in determining water levels. The SPLASH model of Jelesnianski (1972), developed and used to forecast hurricane surges along the east coast of the United States, was applied to the east coast of India by Ghosh (1977). Murty et al. (1986) review these and other

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contributions. The above models were generally used in the Bay of Bengal to run hindcast simulations of past events, typically driven by idealized and symmetric cyclones moving along straight line tracks with constant intensity.

This paper describes a numerical model, intended to provide a basis for improved flood forecasts. The model is used in an investigation of the cyclone surges of April 1991 and November 1970. The present scheme differs from those mentioned above in a number of important respects. Whereas earlier models of the region have either ignored the complex network of linked channels that makes up the Ganges Delta or represented them as a single idealized river, the formulation used here allows the delta to be included in a much more complete and realistic manner than hitherto. This is achieved by a simple modification of the standard depth-averaged storm surge equations to allow them to represent either one-dimensional flow, as in a channel, or two-dimensional flow, as in the open sea. The approach facilitates the design of a model for complex regions like the delta by combining the solution of 1D and 2D equations within the same computational framework, differentiated only by the choice of parameters defining channel width and wetted surface area. A similar approach was used by Oey et al. (1985) to combine 1D and 2D models such that the resulting code would run efficiently on a *vector* processor. They applied their model to the Hudson–Raritan Estuary.

Additionally, the present model allows for inundation using the scheme originated by Flather and Heaps (1975). In combination with the schematization outlined above, this permits 1D flow along a river channel to transform into 2D flow as the water surface elevation rises above the bank level and floods adjacent land areas. As pointed out by a referee, similar developments of the model of Reid and Bodine (1968) were implemented in the United States in the 1970s (Reid et al. 1977) but unfortunately not published in the open literature. That work differs from that described here in a number of ways. First, the model of Reid et al. (1977) neglects advection, the rotation of the earth, and variations within the model of surface atmospheric pressure, all included here. Second, the 1D channels and 2D open-sea flow are treated separately with different formulations, and additional distinct equations give discharge rates over submerged barriers. Here, the basic equations are written in a general form and applied generally as in Oey et al. (1985).

A preliminary version of the present model was used by Flather and Khandker (1993) to investigate possible consequences of rising sea levels on tides, storm surges, and coastal flooding in Bangladesh. In common with earlier work, the present scheme makes use of an analytical model for the surface pressure and wind field distribution in a tropical cyclone. The formulation used here is due to Holland (1980), with data on cyclone position, maximum wind speed, radius to maximum winds, and central pressure kindly provided by the Joint Typhoon Warning Center (hereafter referred to as JTWC). Unlike earlier applications, cyclone evolution is taken into account (by interpolation of the parameters between specified data times or between the satellite fixes from which they were derived) and arbitrary track and cyclone motion are allowed (by computing the speed and direction of transition of the storm from successive positional fixes). The model is described in section 2, and details of the numerical scheme are given in the Appendix.

In section 3 results from model runs are presented. These include a hindcast of the April 1991 storm, and comparisons with available information on flooding extent and estimated water levels. As is usual in severe cyclone floods, it appears that no records survived from the tide gauges in the area, which were either damaged or destroyed. The model results appear to fit the available information very well. Highest water levels in the model hindcast occurred south of Chittagong, where the peak surge coincided closely with tidal high water. Maximum levels in the Meghna Estuary were lower because peak surge occurred shortly after low tide.

Using data from a routine cyclone warning from JTWC (distributed via the meteorological Global Telecommunications Network, and forwarded to the author by Metroute, the ship routing section within the U.K. Meteorological Office), an experimental surge "forecast" was also carried out. Here, the simplest possible assumptions were made and only data available 18–20 hours *before* landfall were employed. The results confirm the critical importance of an accurate knowledge of the movement and evolution of the cyclone for flood forecasting but also suggests that, with further refinement and an appropriate operational system, useful predictions could be made.

Finally, since the April disaster has inevitably been compared with that caused by the great cyclone of November 1970, a hindcast simulation of this latter event was carried out. Estimated maximum water levels were indeed similar in the two events, but the more northerly track of the November 1970 storm produced the highest levels and more widespread flooding in the Meghna estuary and delta region. Levels on the coast south from Chittagong were higher in April 1991.

2. The storm surge forecast model

The mathematical basis of the surge forecast model is described in this section. The numerical scheme used to solve the equations is described in the Appendix.

a. Basic equations

In geographical coordinates, the basic depth-averaged equations governing tide–surge motion and applicable to either 2D motion in an open sea area or to 1D motion along a channel may be written in the form

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$$\frac{\partial \zeta}{\partial t} + \frac{1}{R\cos\varphi}\left\{\frac{1}{B_u}\frac{\partial}{\partial \chi}(B_u D u) + \frac{1}{B_v}\frac{\partial}{\partial \varphi}(B_v D v \cos\varphi)\right\} = 0, \quad (1)$$

$$\frac{\partial u}{\partial t} + \frac{u}{R\cos\varphi}\frac{\partial u}{\partial \chi} + \frac{v}{R}\frac{\partial u}{\partial \varphi} - \frac{uv\tan\varphi}{R} - fv = -\frac{g}{R\cos\varphi}\frac{\partial \zeta}{\partial \chi} - \frac{1}{\rho R\cos\varphi}\frac{\partial p_a}{\partial \chi} + \frac{1}{\rho D}(F_s - F_b) + A_h\nabla^2 u, \quad (2)$$

$$\frac{\partial v}{\partial t} + \frac{u}{R\cos\varphi}\frac{\partial v}{\partial \chi} + \frac{v}{R}\frac{\partial v}{\partial \varphi} + \frac{u^2\tan\varphi}{R} + fu = -\frac{g}{R}\frac{\partial \zeta}{\partial \varphi} - \frac{1}{\rho R}\frac{\partial p_a}{\partial \varphi} + \frac{1}{\rho D}(G_s - G_b) + A_h\nabla^2 v, \quad (3)$$

where

t	denotes time
χ, φ	east longitude and latitude
ζ	elevation of the sea surface
u, v	components of the depth-mean current, $\mathbf{q}$
B_u, B_v	parameters defining channel width for a 1D channel
F_s, G_s	components of $\boldsymbol{\tau}_s$, the wind stress on the sea surface
F_b, G_b	components of $\boldsymbol{\tau}_b$, the bottom stress
p_a	atmospheric pressure on the sea surface
D	the total water depth ($=h+\zeta$, where h is the undisturbed depth)
ρ	the density of sea water, assumed uniform
R	the radius of the earth
g	the acceleration due to gravity
f	the Coriolis parameter ($f = 2\omega\sin\varphi$, where ω is the angular speed of rotation of the earth)
A_h	horizontal eddy viscosity parameter.

Equation (1) is the continuity equation, expressing conservation of volume. In simple terms, considering a vertical column in the sea, changes in surface elevation, ζ, associated with the tide or storm surge, are related to net fluxes of water in or out of the column. Equations (2) and (3), for each coordinate direction, equate the acceleration of the water (on the left side of the equation) to the forces acting on it (on the right). The atmospheric forces, which generate storm surges, are thus wind stress and horizontal gradients of surface atmospheric pressure. Note that, setting $B_u = B_v = 1$, (1) reduces to the standard form of the continuity equation for 2D motion; whereas with $v = 0$, (1) and (2) describe flow in a 1D channel oriented east–west. Similarly, if $u = 0$, (1) and (3) give flow in a north–south oriented 1D channel. In the most general formulation, allowing continuous transformation between 2D and 1D flow, as described in Flather and Hubbert (1990), B_u and B_v would be functions of water level (and hence time). Because of the lack of adequate data on channel configuration in the present application they have been assumed to be constants, equivalent to considering channels of rectangular section.

The problem is closed by relating bottom stress, $\boldsymbol{\tau}_b$, to the depth mean current, $\mathbf{q}$, using a quadratic law

$$\boldsymbol{\tau}_b = k\rho\mathbf{q}|\mathbf{q}|, \quad (4)$$

where k is a friction parameter, and relating the wind stress, $\boldsymbol{\tau}_s$, to the hourly mean surface (10 meter) wind velocity, $\mathbf{w}$, also using a quadratic law

$$\boldsymbol{\tau}_s = c\rho_a\mathbf{w}|\mathbf{w}|, \quad (5)$$

where ρ_a is the density of air and c a drag coefficient. The consensus, based on empirical work over the last 10–15 years is that c depends weakly on wind speed w, reflecting increasing roughness of the sea surface with increasing wind speed. The relationship suggested by Wu (1982) covering conditions ranging from breeze to hurricane has been used here; namely,

$$c \times 10^3 = 0.8 + 0.065w \quad (6)$$

(where w is in m s^{-1}).

Initial and boundary conditions must also be specified. Initial conditions may take the form of prescribed distributions of ζ, u, and v or simply a "cold start" condition

$$\zeta = u = v = 0 \quad \text{at time} \quad t = 0. \quad (7)$$

On coastal boundaries, which are allowed to change position with changing water level according to the method given by Flather and Heaps (1975),

$$q_n = 0, \quad (8)$$

where q_n is the component of current along the outward-directed normal to the boundary. On open sea boundaries a "radiation" condition

$$q_n = \hat{q}_n + \frac{c}{h}(\zeta - \hat{\zeta}) \quad (9)$$

is applied, where $\hat{\zeta}$ and $\hat{q}_n$ are specified functions of position and time, which define the component of tide and surge motion propagating into the area under consideration.

Equations (1)–(3) with (4), (5), and (6) must then be solved within the required domain, subject to conditions (7)–(9), with prescribed atmospheric forcing, and with $\hat{\zeta}$ and $\hat{q}_n$ defined as functions of time and position along its open-sea boundaries. A simple explicit numerical finite-difference method, described in the Appendix, is used to solve the equations.

b. The cyclone model

Meteorological data for the April 1991 cyclone consisted of the following sets of information from JTWC:

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1) Standard "real-time" warnings including, for the specified data time, the estimated position, χ_0, φ_0, of the cyclone center; the maximum 1-min sustained wind speed, w_m; an indication of the extent of strong winds, as in the following example:

radius of over 100-kt winds — 35 n mi

radius of over 50-kt winds

—110 n mi northeast semicircle

90 n mi elsewhere

radius of over 30-kt winds

—250 n mi northeast semicircle

200 n mi elsewhere.

The warnings also give *forecasts,* typically for 12, 24, 48, and 72 hours after the data time, containing similar information. The warnings are issued at intervals of 6 h during the lifetime of the cyclone.

2) "Best track" data and valuable advice, received subsequently from JTWC, giving refined estimates of χ_0, φ_0, and w_m, also at 6-h intervals. Also provided were estimates of the diameter of the eye of the cyclone derived from analysis of satellite images at irregular intervals. It was suggested that the radius of the core of the cyclone be used as an estimate of r_m, the distance from the cyclone center to the radius of maximum winds. Also, following advice from JTWC, the empirical relation due to Atkinson and Holliday (1977)

$$w_m = 6.7(p_n - p_c)^{0.644} \tag{10}$$

was used to estimate the central pressure, p_c (hPa), of the cyclone from w_m (kt). Here p_n is the ambient pressure, taken as 1010 hPa.

The model due to Holland (1980) was used to define the surface atmospheric pressure, p_a, as a function of radial distance, r, from the cyclone center. Thus,

$$p_a = p_c + \Delta p \exp(-A/r^B), \tag{11}$$

where $\Delta p = p_n - p_c$ is the pressure deficit and A and B are empirical constants. The wind profile was calculated from the gradient wind equation

$$\begin{aligned} w_g &= -\tfrac{1}{2}fr + \left\{\frac{1}{4}f^2r^2 + \frac{r}{\rho_a}\frac{\partial p}{\partial r}\right\}^{1/2} \\ &= -\tfrac{1}{2}fr \\ &\quad + \left\{\frac{1}{4}f^2r^2 + \frac{B}{\rho_a}\Delta p(A/r^B)\exp(-A/r^B)\right\}^{1/2}. \end{aligned} \tag{12}$$

In the region of maximum winds, the Coriolis force can be neglected, and (12) reduces to the cyclostrophic wind equation,

$$w_c = \left\{\frac{B}{\rho_a}\Delta p(A/r^B)\exp(-A/r^B)\right\}^{1/2}. \tag{13}$$

The radius to maximum winds, r_m, can then be derived by differentiating (13) w.r.t. r and setting $\partial w_c/\partial r = 0$. This gives

$$r_m = A^{1/B} \tag{14}$$

and, by substituting (14) into (13), the maximum wind speed

$$w_m = (B\Delta p/\rho_a e)^{1/2}. \tag{15}$$

In the present work, A and B were determined by fitting these forms to the data described above. Given w_m, r_m, and Δp, the obvious approach is to calculate B directly from (15) and then A from (14). However, the resulting wind profile was found to be rather sensitive to r_m, the radius of maximum winds, which is itself somewhat uncertain. Since good wind estimates are of primary importance in determining the storm surge, this is not ideal. An alternative approach was therefore adopted, making use of the data on the extent of strong winds, described above, directly. The procedure was an iterative one, using w_m and Δp as given, assuming an initial value for r_m from the satellite information, then using (15), (14), and (12) to compute the radial wind profile. The profile was then compared with the data on the extent of winds exceeding certain speeds, r_m and w_m adjusted and the procedure repeated. This manual adjustment was straightforward and provided, typically after two or three tries, a good fit to the JTWC data with values of A and B consistent with theoretical considerations as given in Holland (1980).

The full procedure for computing the surface wind stress (in E- and N-directed components) and pressure at an arbitrary geographical position χ, φ, and time t, as required in the storm surge computations, was as follows:

1) Use linear interpolation in time to get w_m, Δp, and r_m (adjusted as above) at the required time, t, from appropriate 6-h values. (In the numerical calculations described below, meteorological data were defined at hourly intervals.)

2) Compute A and B using (14) and (15).

3) Calculate r and θ, the radial distance and direction from the cyclone center, (χ_0, φ_0) to (χ, φ).

4) Calculate p_a and the E- and N-directed components of the 1-minute sustained gradient wind vector, $\mathbf{w}_g$, at χ, φ from (11) and (12), and multiply the wind components by a factor (assumed = 0.8) to convert them to hourly mean values. (The direction of the gradient wind is assumed to be $\theta + 90°$, with θ measured clockwise from the north; that is, the cross isobar angle is assumed to be zero.)

5) Add the E- and N-components of storm motion, $\mathbf{V}_s$, computed from successive 6-h positions, to the corresponding components of hourly mean wind. This

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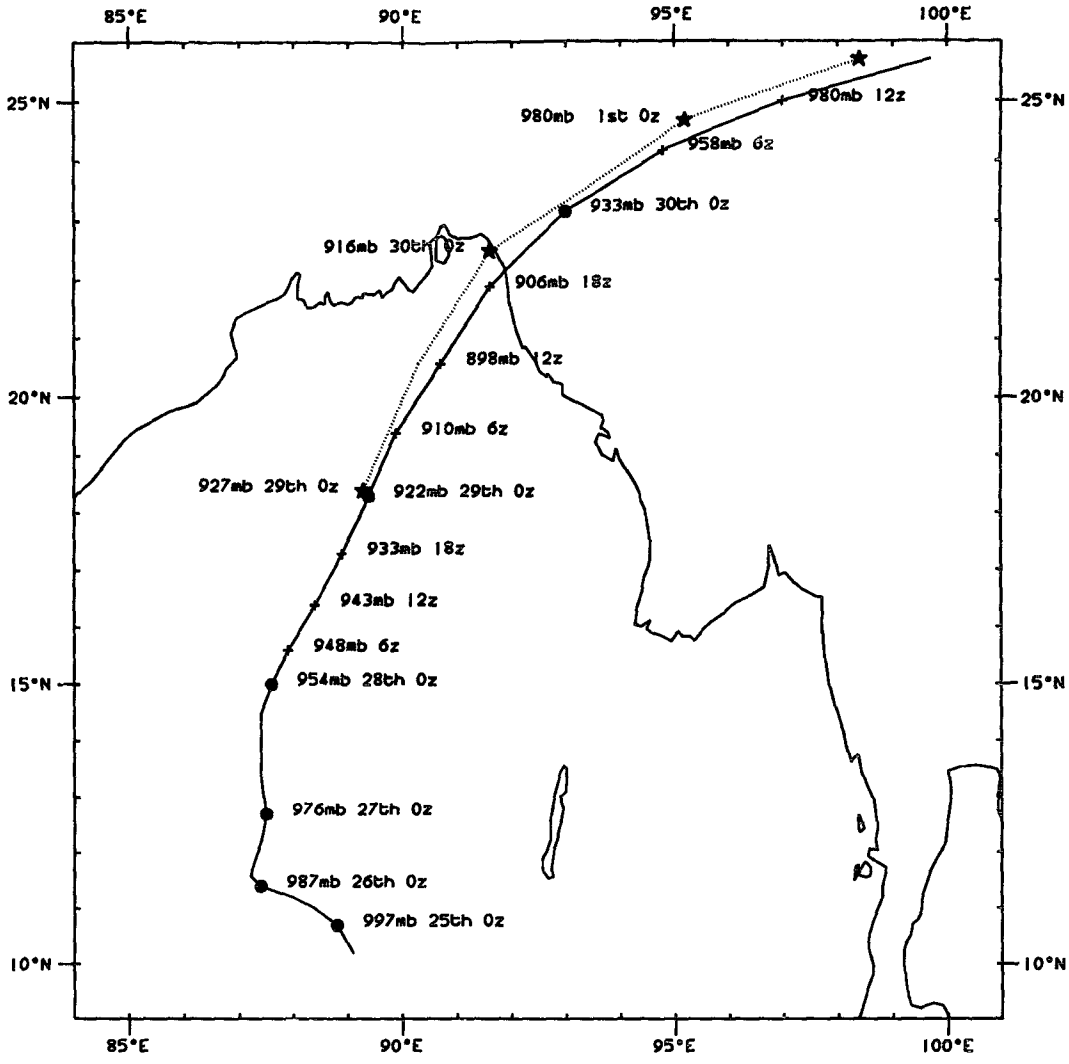

FIG. 2. Track and central pressure of cyclone "O2B," April 1991, from JTWC "best track" data (continuous line). Also shown (dotted line) is the predicted track and central pressure from the JTWC cyclone warning issued at 0000 UTC 29 April 1991.

introduces an asymmetry into the model cyclone wind field.

6) Compute the wind stress, in components, using (5) and (6).

It should be pointed out that this cyclone model is somewhat simpler than those used, for example, by the U.S. National Hurricane Center for North Atlantic hurricanes and for cyclones affecting Australian coasts (Hubbert et al. 1991). Hubbert et al., although using the basic Holland model, adopt a somewhat different approach, including a nonzero value for the cross-isobar angle. The present model could certainly be improved and alternatives should be examined in the future. Eventually, the required atmospheric fields might be generated from a numerical atmospheric boundary-layer model of appropriate high resolution, much as regional atmospheric model forecasts are now used to drive models for predicting storm surges due to extratropical depressions (e.g., Flather 1991).

c. *Application to the northern Bay of Bengal*

The computational grid of the numerical model is shown in Fig. 1. It covers the region north of 20°N with a grid size of 1/10° by 1/10°, equivalent to about 10 km (note that the variable channel widths in the delta region are not reproduced in this diagram). To allow for inundation, the computational area includes low lying coastal land delimited by the broken line. On the southern open sea boundary and at the upstream limit of the major rivers, input disturbances comprising four tidal constituents (M_2, S_2, K_1, and O_1) and a surge component derived from the hydro-

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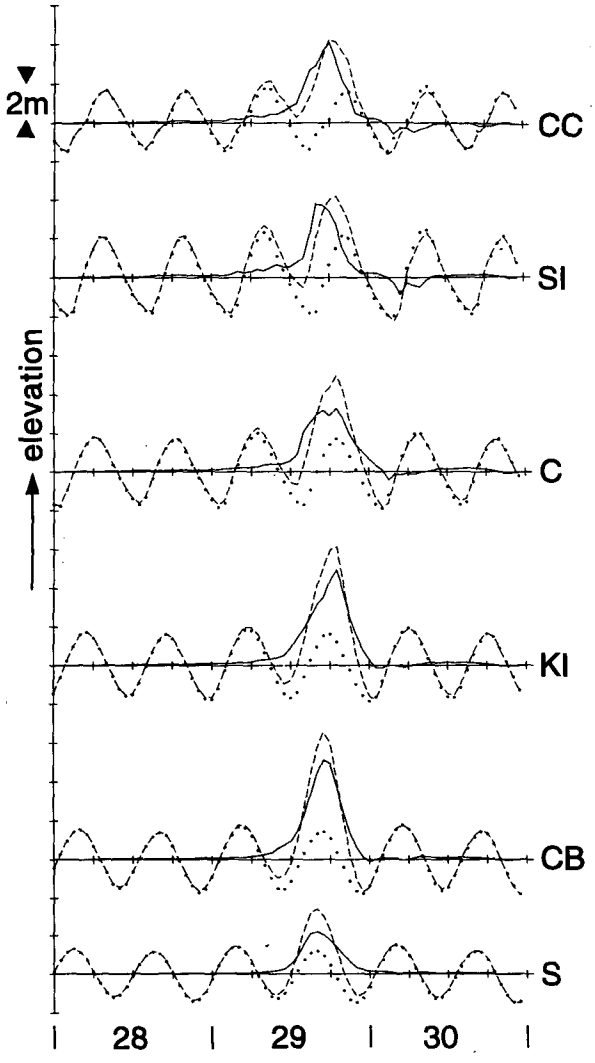

FIG. 3. Variation with time of the computed total water level (broken line), tidal elevation (dotted line), and the storm surge residual (continuous line) for the period 0000 UTC 28 April to 0000 UTC 1 May 1991 from the hindcast model simulation for the locations shown in Fig. 1b (CC: Char Changa, SI: Sandwip Island, C: Chittagong, KI: Kutubdia Island, CB: Cox's Bazar, and S: Shapuri Island).

static law were introduced. From results of preliminary tidal calculations, the bottom friction parameter was taken as $k = 0.0015$, somewhat lower than is usual for North Sea models, for example. The horizontal eddy viscosity was taken to be $A_h = 100\ \mathrm{m^2\ s^{-1}}$.

3. The cyclone surge of 29–30 April 1991

In this section, we describe the application of the model to an investigation of the storm surge generated by the cyclone that struck Bangladesh on 29–30 April 1991.

a. Hindcast simulation

Figure 2 shows the track (continuous line) and central pressure (in mb; 1 mb = 1 hPa) of the cyclone "O2B" based on the JTWC "best track" data. It developed from a tropical disturbance, reaching cyclone strength late on 24 April in the central southern part of the bay. After moving slowly west-northwest for about a day, it turned north, continuing to organize and intensify. By 0000 UTC 28 April, the maximum (1 min sustained) wind speed was about 90 kt (46.3 m s^{-1}) and central pressure 954 hPa [from (10)]. From about this time, the cyclone turned northeast toward the Meghna Estuary and accelerated. It continued to intensify, the estimated maximum wind speed reaching 140 kt (72 m s^{-1}) by 1200 UTC 29 April, with corresponding central pressure about 898 hPa. Landfall on the coast of Bangladesh north of Chittagong was at about 2000 UTC 29 April (0200 local time on 30 April 1991).

Using the "best track" data, a model hindcast simulation covering the period from 0000 UTC 27 April to 0000 UTC 1 May 1991 was carried out. The model was run twice; first with tidal input and atmospheric forcing simulating the total tide and surge motion, and second, without atmospheric forcing, to predict the normal tides and allow the surge component to be derived.

Figure 3 shows time series of the total water level (relative to MSL), the normal tidal elevation (also relative to MSL) and the storm surge elevation, defined at any time as the difference between total water level and predicted tide. The series are plotted for Char Changa and Sandwip Island in the Meghna Estuary (points 1 and 2, respectively, in Fig. 1) and for Chittagong, Kutubdia Island, Cox's Bazar, and Shapuri Island (points 3, 4, 5, and 6) along the coast. It can be seen from Fig. 3 that the rapidly moving cyclone produced a surge of quite short duration. For Chittagong and locations to the south, the surge peak coincided closely with high tide, so that the effects were superimposed leading to the high water levels in this area. At Char Changa and Sandwip, the surge peak preceded tidal high water by about 3 h and the surge elevation decreased to less than half its maximum value before high tide. Consequently total water levels were substantially lower.

Figure 4a shows the surface pressure field and wind vectors (plotted for every second grid point of the model within the computational domain) at 1800 UTC 29 April, shortly before landfall. The corresponding distributions of storm surge elevation and depth mean current from the model hindcast are shown in Figs. 4b and 4c. The strongest onshore winds affected the mainland coast between Kutubdia Island and Cox's Bazar and produced the highest surges at about this time. North of 22°N, the winds are directed toward the west, that is, offshore, reducing the surge at Chittagong but producing a complex surge distribution among the islands in the delta. The pattern of surge currents (Fig. 4c) is interesting, containing two circulation cells. These are related to the shelf bathymetry, which has an inner shallow region

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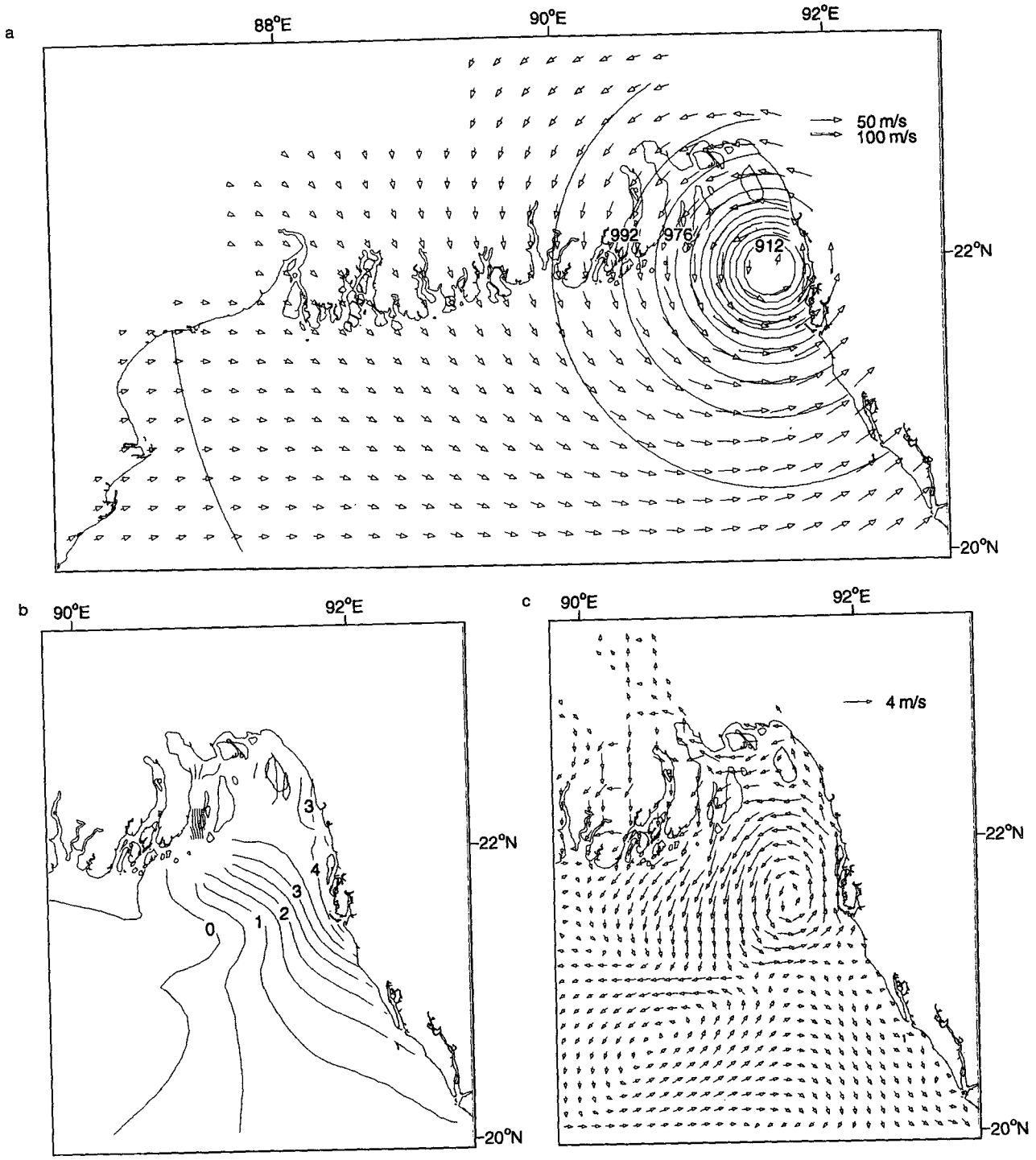

FIG. 4. Model fields at 1800 UTC 29 April 1991, shortly before landfall, from the hindcast simulation. (a) Surface pressure contours and wind vectors (plotted for every second grid point); (b) storm surge residual elevation (m) and (c) storm surge depth mean current vectors. [Panels (b) and (c) show distributions only where there are significant variations in the eastern half of the model.]

(<20 m deep), a marked step (20 to 100 m), and a broad outer region delimited by the shelf slope to the south and a canyon, the "Swatch of No Ground," to the west (see Fig. 1). Clearly, there is strong input of vorticity as the cyclone traverses the shelf which combined with this bathymetry produces the circulation shown. The pattern of flow persists for some hours after the cyclone has passed.

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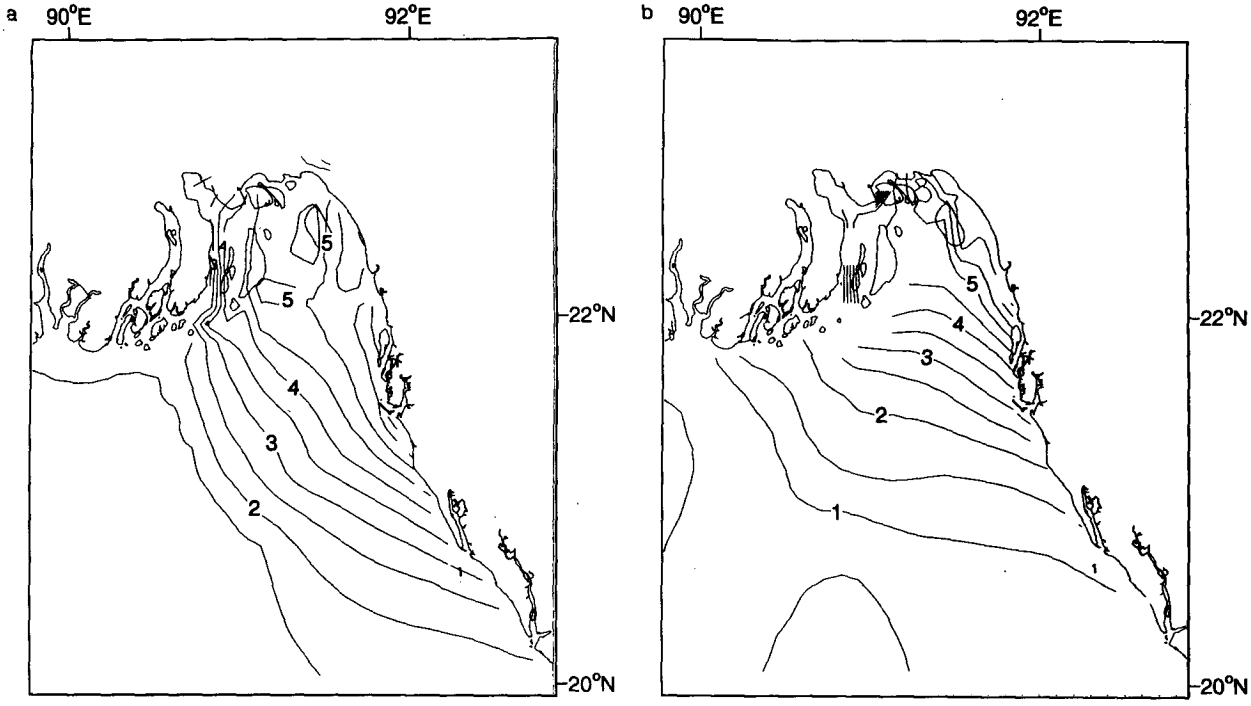

FIG. 5. Distributions of (a) maximum computed water level (m) due to tide and surge combined, and (b) maximum storm surge elevation (m), in the period 0000 UTC 28 April to 0000 UTC 1 May 1991, from the model hindcast simulation.

The distributions of maximum computed water level and maximum surge elevation during the storm are shown in Fig. 5. The highest water level was 6.59 m above mean sea level (MSL) at Cox's Bazar with levels only slightly lower on the coast near Kutubdia Island. In the Meghna Estuary, maximum levels exceeded 5 m only on the east coasts of Sandwip and Hatia Islands. The largest surge elevation, 5.20 m occurred in the Meghna estuary NW of Sandwip, with the next highest value, 5.12 m, at Cox's Bazar. The coast near Kutubdia Island was affected by surges reaching 4.6 to 4.9 m. The influence of reduced atmospheric pressure raising surge levels by 0.5 to 1.0 m, as predicted by the hydrostatic law, is apparent in the SW corner of Fig. 5b, where the cyclone track crosses the outer shelf. Over the shallow inner shelf wind-stress forcing predominates. According to the model solution, the islands in the Meghna Estuary and most low-lying coastal land in the area bounded by 90°45′E and 92°E and south of 23°N were flooded. The surge also penetrated for some distance into the channels of the eastern delta.

As stated earlier, data for direct verification of the model hindcast results is very limited. The Bangladesh Inland Water Authority, which maintains an extensive network of tide gauges, reported that no data for the cyclone period had been recovered. However, they kindly provided a map, reproduced in Fig. 6, showing the areas affected by the storm. This is in good agreement with the model results, Fig. 5a. A report on the cyclone disaster and subsequent relief effort (G. P. Sevenhuysen 1991) also contains useful verification, in particular giving the time of flooding as "between 12 and 1 o'clock in the morning of 30 April" (local time; equivalent to 1800 to 1900 UTC 29 April), and listing the areas flooded, which agree well with the extent of flooding indicated by Fig. 5a.

To summarize, it appears that the model reproduced the water levels and flooding during the event with good accuracy. The coincidence of peak surge and tidal high water and the rapid movement of the cyclone, giving a large surge of short duration, were key factors in producing the highest levels along the Chittagong–Cox's Bazar coast. The flooding occurred in the early hours of 30 April (local time) and hence in darkness, which must have increased the number of casualties.

b. An experimental forecast

To examine the problems of forecasting the cyclone-generated storm surge, a further model simulation was carried out. Anticipating that information provided 12 to 18 h before the event could be most useful, possibly allowing evacuation of threatened areas in daylight, the forecast was based on the warning issued by JTWC at 0000 UTC 29 April. The predicted track and cyclone development is shown (as a dotted line) in Fig. 2. By comparison with the "best track" data, it can be seen that the prediction is remarkably accurate, with the

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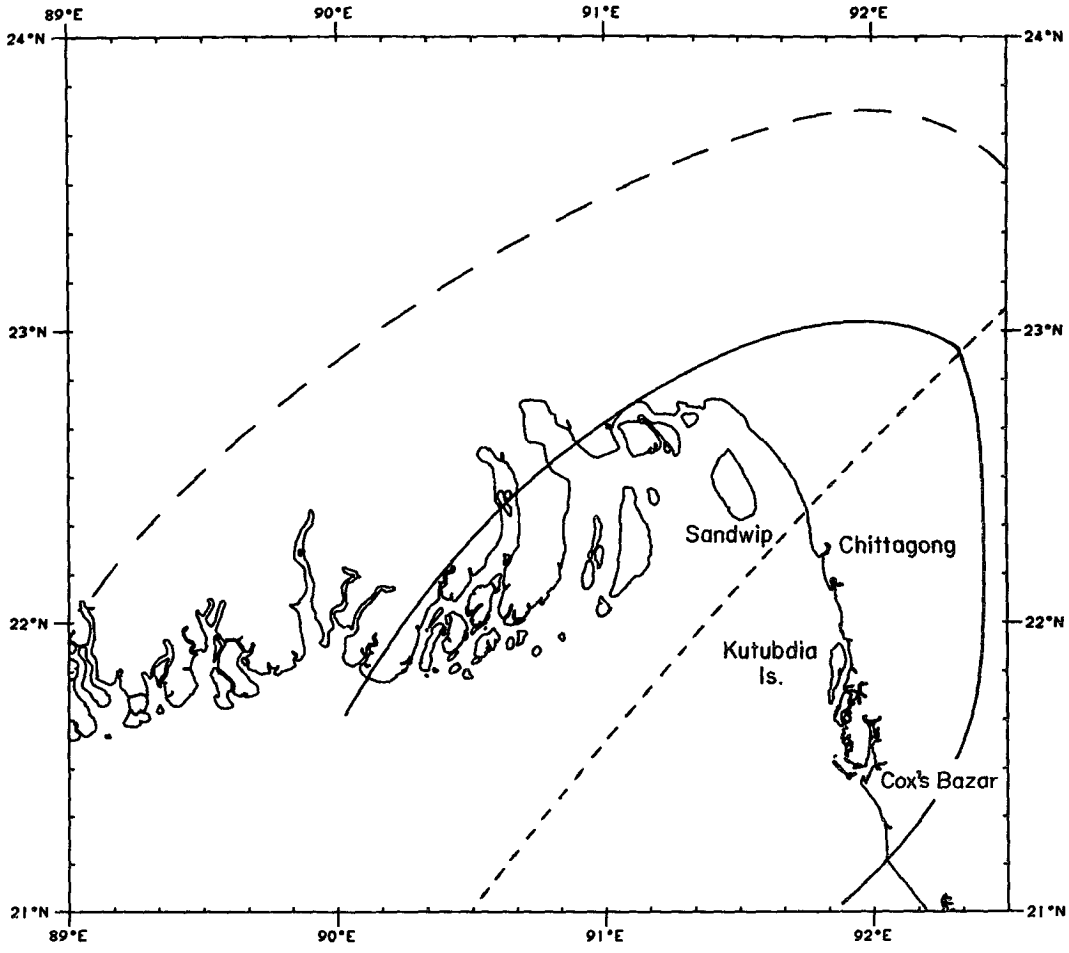

FIG. 6. Plot of region affected by the cyclone of April 1991; the area most seriously affected is bounded by the continuous line, with less serious effects to the broken line. (Redrawn from information provided by the Bangladesh Inland Water Transport Authority.)

expected landfall point only 50 km north of the actual location and the maximum wind speed only 10% low. However, the forecast could not anticipate the acceleration of the cyclone during the 12 h preceding landfall, so that the predicted landfall time was about 5 h late.

Figure 7 shows time series of the total water level, tidal elevation, and storm surge elevation, for the same locations as Fig. 3. Figure 8 shows the distributions of maximum computed water level and maximum surge elevation from the simulated forecast run, which may be compared with corresponding results from the hindcast in Fig. 5. Also plotted are distributions of "forecast error," computed by taking differences between the forecast and hindcast values of maximum computed water level (Fig. 8c) and maximum computed surge elevation (Fig. 8d).

Consider first forecast errors in the surge component. Because of the slower movement of the cyclone along its track, severe winds affected the coastal region for a significantly longer period, producing a larger surge with increased duration than in the hindcast (cf. Figs. 7 and 3). The more northerly track of the cyclone in the forecast was an additional factor, shifting the largest surges toward the Meghna Estuary (see Figs. 8b and 5b). Forecast errors in maximum surge elevation (Fig. 8d) exceeded +2.5 m along a 50-km stretch of coast centered on Chittagong, the area just south of the actual landfall. This corresponds to the coastal area where the error in wind speed and direction was greatest. Farther south, forecast surge maxima were underpredicted by up to 1.5 m, reflecting the northerly shift. Between these two extremes, in fact at Kutubdia Island, the maximum surge elevation was correct, but about 5 h late (cf. Figs. 3 and 7).

The later time of landfall gave a surge peak 3 to 5 h after tidal high water on the Chittagong–Cox's Bazar coast, and as a result maximum water levels were underpredicted by more than 1.5 m (Fig. 8c). At Kutubdia Island and Cox's Bazar the errors were, respectively,

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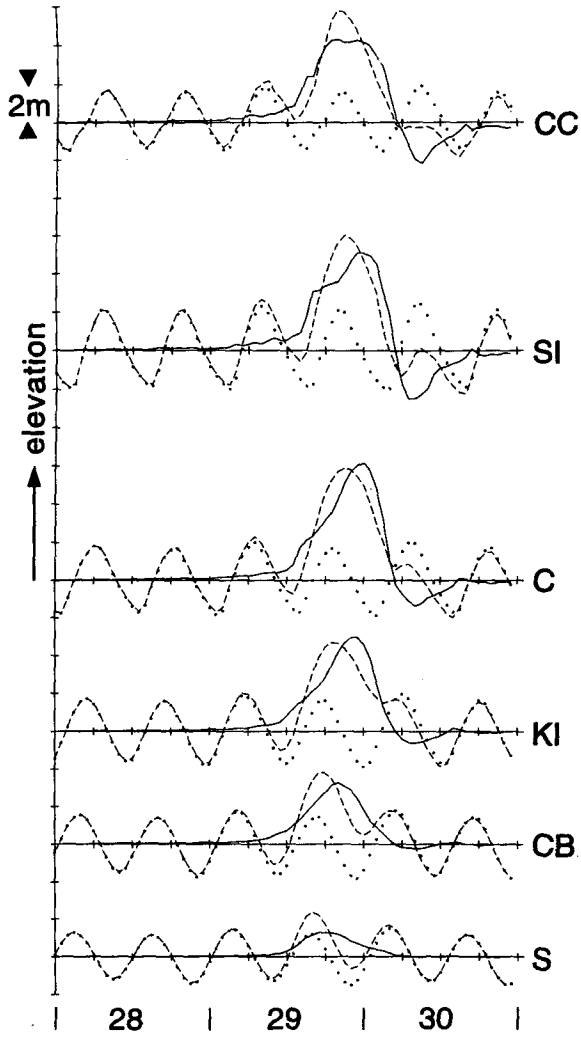

FIG. 7. Variation with time of the computed total water level (broken line), tidal elevation (dotted line), and the storm surge residual (continuous line) for the period 0000 UTC 28 April to 0000 UTC 1 May 1991 from the model *forecast* simulation. Locations as in Fig. 3.

−1.5 and −1.8 m. The time of maximum water level was, however, predicted quite accurately in this area. At Chittagong and in the Meghna Estuary, the surge was delayed and enhanced; the southward flow of the ebbing tide being held back by the strong winds from the south. The surge thus extended over high tide, giving the highest water level, 7.21 m, on the Noakhali coast north of Sandwip Island. Maximum computed levels were generally overpredicted by 1 to 2 m in this area; at Chittagong the forecast error was +0.9 m. Maximum surges exceeded 6.0 m in quite a large area east and north of Sandwip. Although there are important differences between hindcast and forecast surges and water levels, as indicated above, the predicted extent of flooding is rather similar.

The preceding analysis of forecast results and comparisons with the hindcast simulation demonstrate the importance of accurate predictions of cyclone track, cyclone development, and, particularly in an area with large tidal range as here, of landfall time. Considering their preliminary nature, the results are very promising. Further experiments to clarify the relative importance of these and other factors are required, and should contribute to the development of a scheme capable of producing useful forecasts for warning purposes.

c. *Comparison with the great storm of November 1970*

Finally, since the cyclone in April 1991 has inevitably been compared with the great cyclone disaster in November 1970, it is of interest to run a simulation of this latter event to examine how the storm surge and water levels differ.

Meteorological data for the November 1970 storm were extracted from various publications and the approach described above adapted to suit the data available. The cyclone track was taken from Frank and Hussain (1971). No information on the variation with time of the maximum wind speed was available and it was therefore assumed that the maximum wind speed was 120 kt (61.8 m s^{-1} = 222 km h^{-1}) throughout, as given by Murty et al. (1986). This is slightly weaker than the estimated maximum wind speed during April 1991. For consistency with the treatment of the April 1991 storm, the pressure deficit was derived using the Atkinson and Holliday (1977) formula, (10), giving $p_n - p_c$ = 88.3 hPa, and hence p_c = 921.7 hPa. The radius to maximum winds was taken to be 30 km (Das 1972).

Figure 9 shows the cyclone track, with cyclone positions defined at 12-h intervals. Comparing this with Fig. 2, it can be seen that the track was farther west and north than that of the April 1991 cyclone, taking the storm center across the southern edge of the delta region between 1200 UTC 12 and 0000 UTC 13 November. The cyclone crossed the mainland coast NE of Sandwip Island at about 2300 UTC 12 November, consistent with observations of maximum gusts at Chittagong reported by Frank and Hussain (1971).

A model hindcast was carried out covering the period from 0000 UTC 10 November to 0000 UTC 14 November 1970. As for the April 1991 event, the model was run twice; first for tide and surge combined and second for tide only. Figure 10 shows plots of time series of computed water level, tide, and storm surge elevation at the six locations used earlier. Because the cyclone took several hours to cross the delta, it was inevitable that peak surge and high tide would occur together at some locations. Frank and Hussain (1971) and Das (1972) suggest that the peak surge and high tide at Chittagong occurred together at about 0000 UTC 13 November. In the present simulation, the peak

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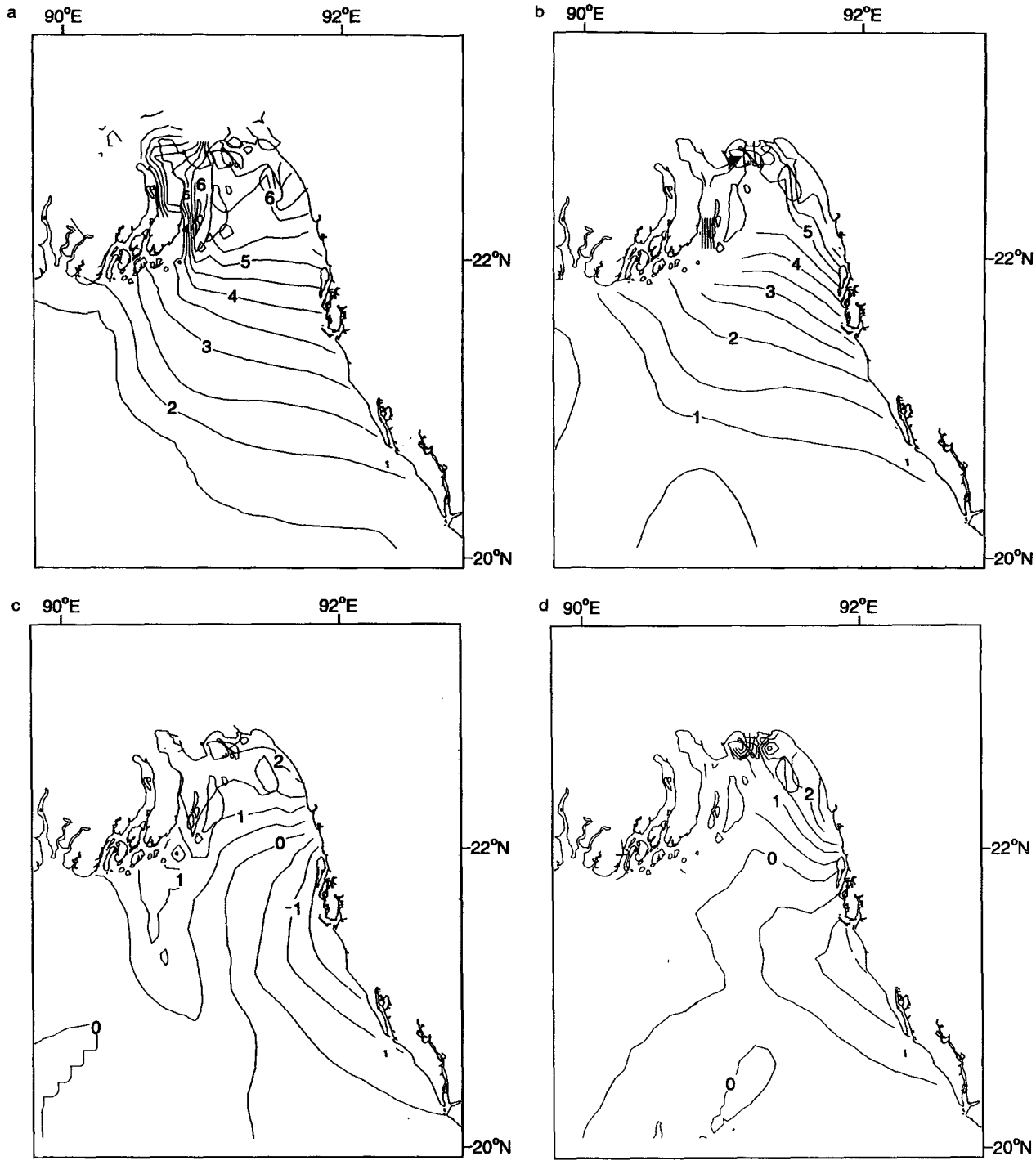

FIG. 8. Distributions of (a) maximum computed water level (m) due to tide and surge combined, and (b) maximum storm surge elevation (m) in the period 0000 UTC 28 April to 0000 UTC 1 May 1991, from the model *forecast* simulation, and (c) forecast error (m) in maximum computed water level, and (d) forecast error (m) in maximum storm surge elevation, defined as differences between forecast and hindcast values.

surge at Chittagong occurs about 1 h earlier, at 2300 UTC 12 November, but tidal high water occurs some 6 h earlier at about 1800 UTC. The relevant predicted high tide (for Chittagong–Karnaphuli River from the Admiralty Tide Tables, Vol. 2, 1970) was at 0025 local time on 13 November; 1825 UTC 12 November, con-

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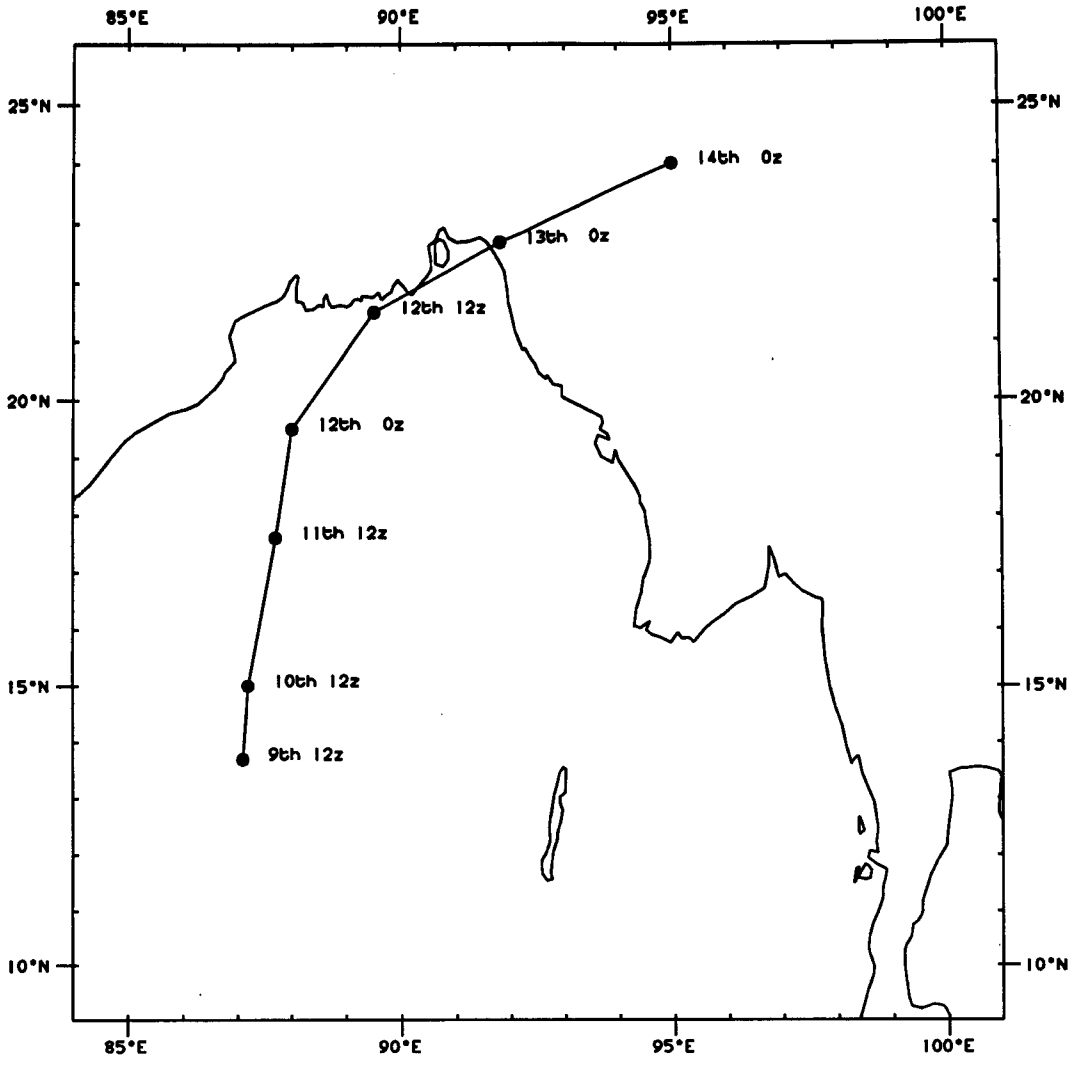

FIG. 9. Track of the November 1970 cyclone, based on Frank and Hussain (1971).

sistent with the model result. It seems, therefore that the peak surge at Chittagong (and at other locations along the coast to the south) occurred shortly before low tide. Peak surge and high tide occurred together in the western parts of the delta, affected some hours earlier by the cyclone winds, and also in the inner Meghna estuary, where high tide occurs later than at Chittagong.

Figure 11 shows the distributions of maximum computed water level and of maximum computed surge elevation. As pointed out earlier, increased surge elevations associated with low atmospheric pressure along the approach track of the cyclone can be seen clearly in Fig. 11b. Both the highest water level, 7.02 m, and the largest surge elevation, 6.13 m, occurred on the Noakhali coast north of Sandwip Island. Large surges and high water levels extended much farther west than in April 1991 (cf. Fig. 5), and according to the model solution, most of the delta was flooded. The most serious effects of the April 1991 and November 1970 cyclones were, therefore, in different areas, though the maximum surge elevations and water levels were similar in magnitude.

4. Conclusions and future prospects

A new storm surge forecast model has been described and applied to the northern Bay of Bengal. Extending numerical schemes used successfully for about 15 years for storm surge prediction in the North Sea (Flather et al. 1991), the new formulation combines the solution of 1D and 2D equations within the same computational framework, differentiated only by factors defining channel width and wetted surface area. This allows models including open sea areas and river channels, such as occur in the delta region of Bangladesh, to be

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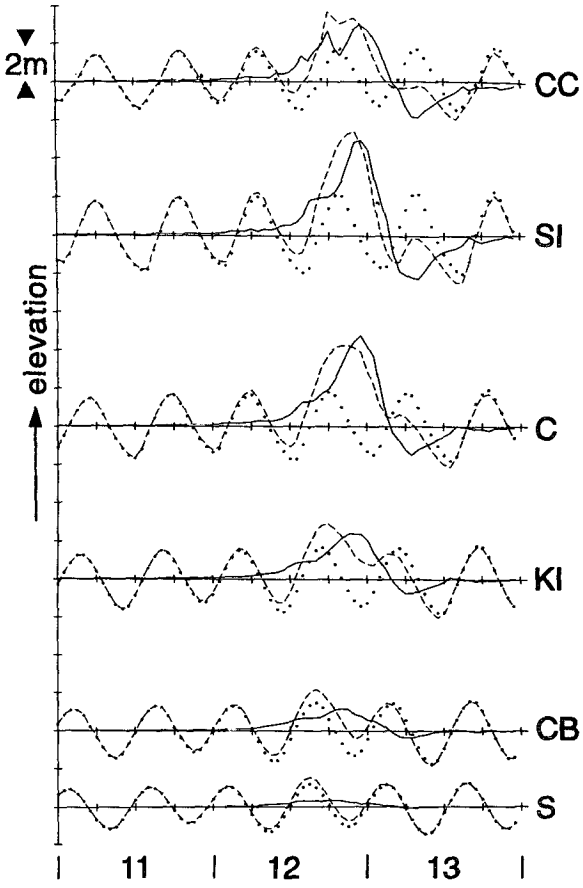

FIG. 10. Variation with time of the computed total water level (broken line), tidal elevation (dotted line), and the storm surge residual (continuous line) for the period 0000 UTC 11 to 0000 UTC 13 November 1970 from the hindcast model simulation.

constructed relatively easily. Further, by allowing for inundation using the simple algorithm introduced by Flather and Heaps (1975), simulation of the transformation of 1D flow along a river channel into 2D flow as the surface elevation rises above bank level and floods adjacent land areas is possible.

The model was used to simulate the storm surge caused by the cyclone in April 1991. A hindcast run based on "best track" data from JTWC reproduced the storm surge and extent of flooding quite well, as far as these are known. The rapidly moving cyclone produced a large surge of short duration, which coincided with high tide on the coast between Chittagong and Cox's Bazar to produce high water levels there. Levels in the delta were generally less extreme because surge levels decreased before tidal high water.

An experimental model "forecast," the first such attempted, was carried out using data from a cyclone warning issued by JTWC 12 to 18 h before landfall. This was a preliminary investigation of the problems associated with predicting cyclone surges in the area, intended to provide an indication of the improvements required to produce useful forecasts. The predicted cyclone track was very accurate, but the forecast landfall time was about 5 h late due to the acceleration of the storm as it approached land. As a result, the storm surge occurred later than in the hindcast and so coincided with high tide in the Meghna Estuary, rather than along the mainland coast.

A simulation of the great storm in November 1970 provided the means of comparing this and the April 1991 events. An important difference was that the track of the November 1970 cyclone was west and north of that in April 1991, taking the storm across the southern delta. The worst effects, therefore, occurred in different areas, though maximum water levels and surge elevations were similar in magnitude. The investigation revealed some confusion in the relative timing of surge and high tide in published accounts of the 1970 event.

Further work on a number of aspects of the problem would clearly be of value. More detailed information on bathymetry and land elevations in the vulnerable areas would allow the simulation of inundation in the tide–surge model to be improved. Thus, the channel width parameters and wetted surface area of grid boxes would depend on water depth and hence time, as proposed by Flather and Hubbert (1990), resulting in better treatment of subgrid-scale processes. The basic model grid should also be refined, perhaps by a factor of three in the NE of the bay, providing a better representation (and better results) in the important nearshore regions.

The comparison of forecast and hindcast simulations of the April 1991 event has indicated the significance of accurate predictions of cyclone track, cyclone development, and, particularly in an area with large tidal range as here, of landfall time. Considering their preliminary nature, the results are very promising, but further experiments to clarify the relative importance of these and other factors (e.g., river flow) are required, and should contribute to the development of a scheme capable of producing useful forecasts for warning purposes. Any improvement in the accuracy of cyclone predictions would clearly be of immediate benefit.

Finally, the design of operational systems should be considered. Whereas the experimental prediction described here was based on a single set of forecast information well ahead of the event, further information would become available at subsequent times. It would be useful to examine the change in forecast accuracy resulting from the use of later cyclone predictions and with alternative assumptions. For example, forecast runs could be carried out every 6 h using data from successive JTWC warnings. It might be expected that, as the time of landfall approaches and later satellite fixes are obtained, the accuracy of the predicted cyclone track, movement, and intensity would improve, leading to better flood forecasts with decreasing lead time. Further, the surge model could be run for different scenarios covering the range of

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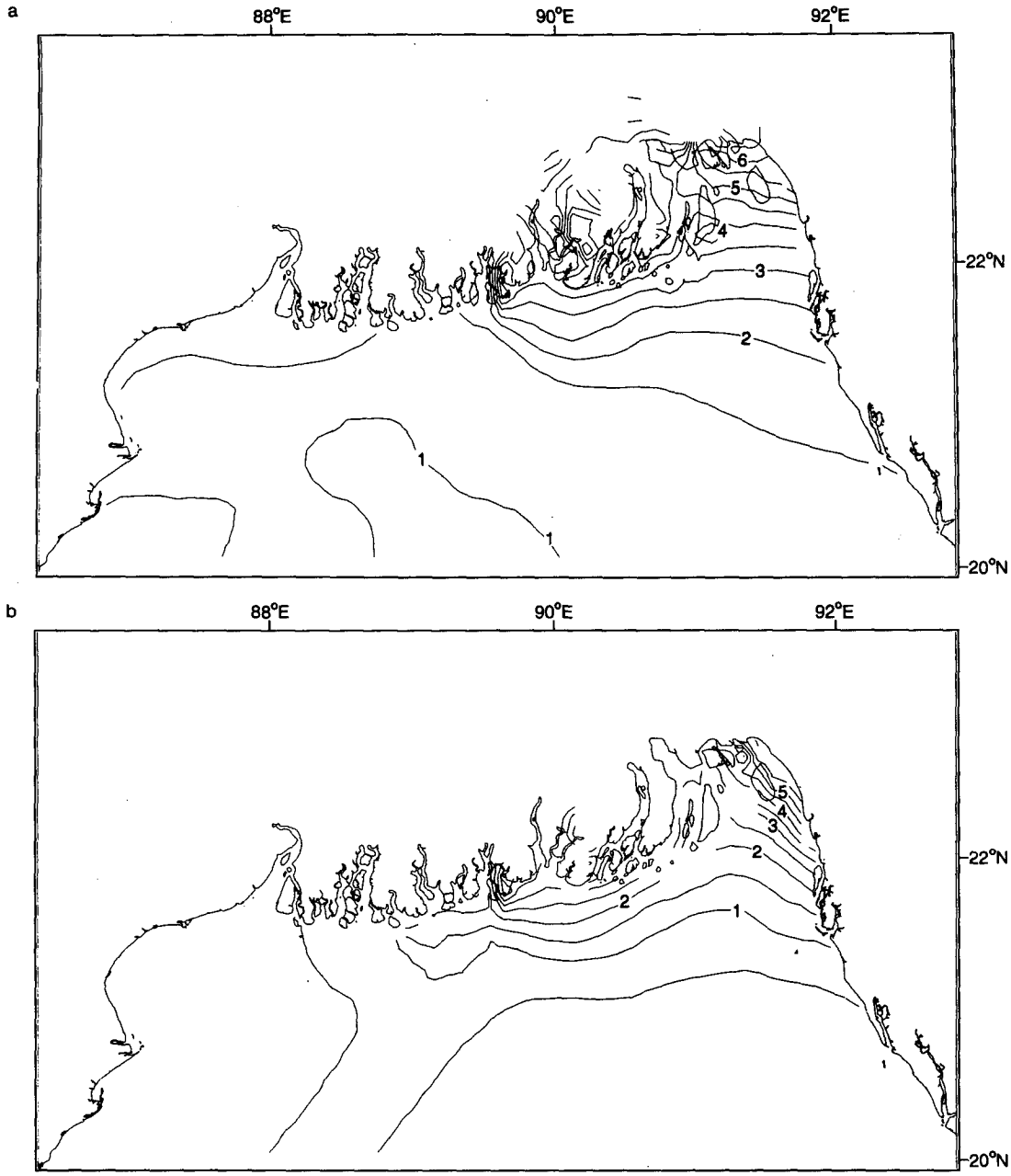

FIG. 11. Distributions of (a) maximum computed water level (m) due to tide and surge combined, and (b) maximum storm surge elevation (m), in the period 0000 UTC 11 to 0000 UTC 13 November 1970, from the model hindcast simulation.

uncertainty in cyclone behavior at the initial time of a forecast. The aim would be to establish an operational procedure in which the forecasts "home in" on the eventual "true" solution. The overall objective would be to produce forecasts of useful accuracy early enough for warnings to be issued in time to allow people to reach a place of safety.

Acknowledgments. Thanks are due to Frank Wells of the Joint Typhoon Warning Center, Guam, for providing cyclone data and valuable advice; to John Pinkau of Metroute, U.K. Met. Office, for forwarding the JTWC bulletins; and to Hasina Khandker of the Bangladesh Inland Water Transport Authority for supplying information on damage caused by the

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April cyclone. Robert Smith helped finalize the diagrams.

APPENDIX

The Numerical Scheme

The numerical scheme adopted for the model is based on and includes elements of those described by Flather and Heaps (1975), Davies and Flather (1978), Proctor and Flather (1983), and Flather and Hubbert (1990). The finite-difference grid used is the Arakawa "*C*-grid" shown in Fig. A1. It consists of a rectangular array of l rows and n columns, with grid spacing (between like points) $\Delta\varphi$ in latitude and $\Delta\chi$ in longitude. "Elements," each consisting of a ζ point (at say χ, φ) denoted by O, a u point (at $\chi + \Delta\chi/2$, φ) denoted by +, and a v point (at χ, $\varphi - \Delta\varphi/2$) denoted by $\times$, are numbered consecutively:

$$i = 1, 2, \cdots, n, n+1, n+2, \cdots, 2n, \cdots (l-1)n+1, \cdots, ln$$

counting by elements from left (W) to right (E) along each row, and moving down the grid row by row (from N to S). Discrete values of the variables at appropriate

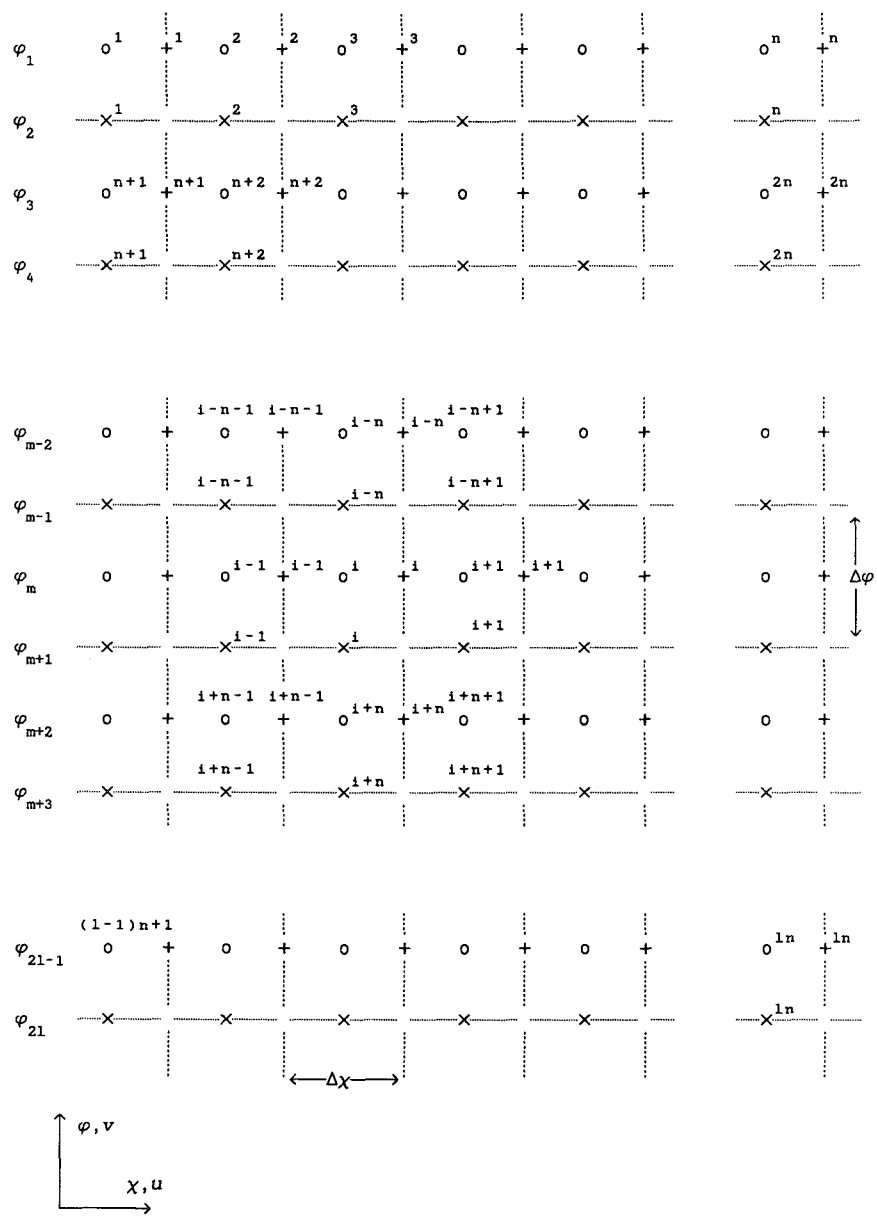

FIG. A1. Finite-difference grid and point numbering system.

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grid points and times are identified by subscripts and superscripts as follows

$$\zeta = \zeta_i^t, \quad h = h_i, \quad D = D_i^t, \quad A = A_i \quad \text{at } \zeta \text{ point } i \text{ and time } t,$$

$$u = u_i^t, \quad B_u = B_i \quad \text{at } u \text{ point } i \text{ and time } t,$$

$$v = v_i^t, \quad B_v = C_i \quad \text{at } v \text{ point } i \text{ and time } t.$$

Following Flather and Hubbert (1990), the continuity equation (1) is approximated by

$$\frac{A_i}{\Delta t}\{\zeta_i^{t+\Delta t} - \zeta_i^t\} = -\frac{1}{R\cos\varphi_m\Delta\chi}\{d_i^t B_i u_i^t - d_{i-1}^t B_{i-1} u_{i-1}^t\} + \frac{1}{R\cos\varphi_m\Delta\varphi} \times \{e_{i-n}^t C_{i-n}\cos\varphi_{m-1} v_{i-n}^t - e_i^t C_i \cos\varphi_m v_i^t\}, \quad \text{(A1)}$$

where

- Δt the time step,
- d_i^t $0.5(D_i^t + D_{i+1}^t)$, the total water depth at u point i,
- e_i^t $0.5(D_i^t + D_{i+n}^t)$, the total water depth at v point i,
- A_i "area" factor for element i surface area of element $i/R^2\cos\varphi_m\Delta\chi\Delta\varphi$,
- B_i "breadth" factor for u point i, width of channel at u point $i/R\Delta\varphi$
- C_i "breadth" factor for v point i, width of channel at v point $i/R\cos\varphi_{m+1}\Delta\chi$.

Note that with $A_i = B_i = B_{i-1} = C_i = C_{i-n} = 1$, (A1) reduces to the standard finite-difference form of the continuity equation as given by Davies and Flather (1978). When A_i, B_i, and C_i take other values (which may be > or <1), then according to their value, (A1) may specify conservation of volume for, for example, a narrow channel or a broad flood plain. In these circumstances, the accuracy of the finite-difference approximation of spatial derivatives is clearly reduced (to first order), but this is less important than the failure to correctly define the cross-sectional area of flow along channels. Figure A2 indicates the use of area and breadth factors to approximate channel systems.

The finite-difference form of the equations of motion (1) and (2) follows Proctor and Flather (1983). Thus, (1) is approximated by

$$\frac{1}{\Delta t}\{u_i^{t+\Delta t} - u_i^t\} - 2\omega\sin\varphi_m\tilde{v}_i^t + \frac{1}{4R\cos\varphi_m\Delta\chi} \times [\{u_{i+1}^t\}^2 - \{u_{i-1}^t\}^2] + \frac{1}{4R\Delta\varphi}[\{v_{i-n}^t + v_{i-n+1}^t\}\{u_{i-n}^t - u_i^t\} + \{v_i^t + v_{i+1}^t\}\{u_i^t - u_{i+n}^t\}]$$

$$= \frac{g}{R\cos\varphi_m\Delta\chi}\{\zeta_{i+1}^{t+\Delta t} - \zeta_i^{t+\Delta t}\} - \frac{k}{d_i^t}u_i^{t+\Delta t}[\{u_i^t\}^2 + \{\tilde{v}_i^t\}^2]^{1/2} + \frac{1}{\rho}\left\{-P_i^t + \frac{F_i^t}{d_i^t}\right\} + \frac{A_h}{R^2\cos^2\varphi_m\Delta\chi^2}[\{u_{i+1}^t - 2u_i^t + u_{i-1}^t\}] + \frac{A_h}{R^2\cos\varphi_m\Delta\varphi^2}[\cos\varphi_{m-1}\{u_{i-n}^t - u_i^t\} + \cos\varphi_{m+1}\{u_i^t - u_{i+n}^t\}] \quad \text{(A2)}$$

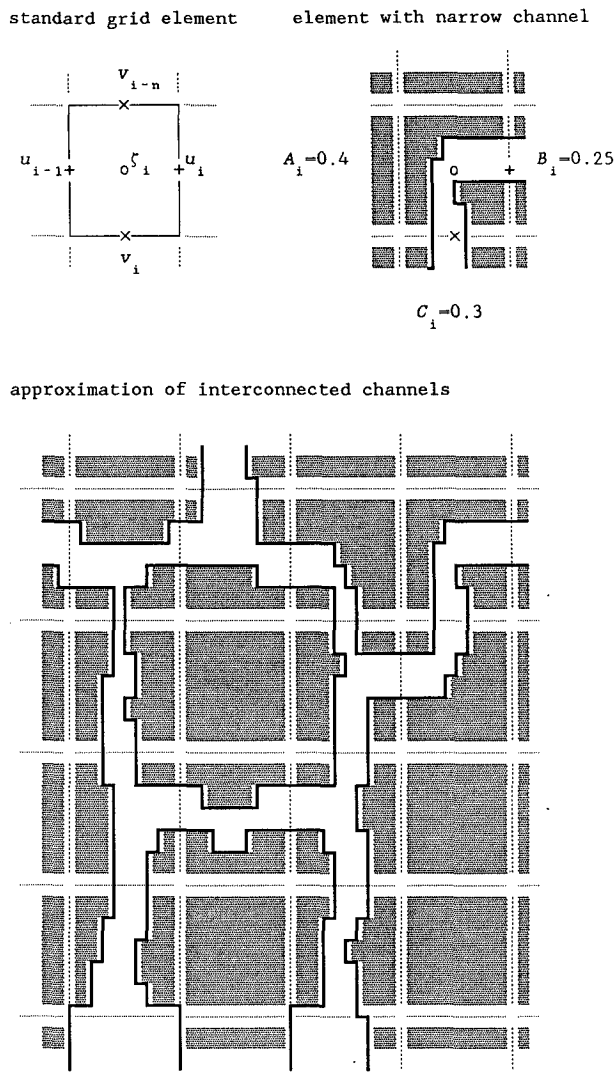

FIG. A2. Use of area and breadth factors to represent narrow channels.

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and (2) by

$$\frac{1}{\Delta t}\{v_i^{t+\Delta t} - v_i^t\} + 2\omega \sin\varphi_{m+1}\tilde{u}_i^{t+\Delta t}$$
$$+ \frac{1}{4R\Delta\varphi}[\{v_{i-n}^t\}^2 - \{v_{i+n}^t\}^2]$$
$$+ \frac{1}{4R\cos\varphi_{m+1}\Delta\chi}[\{u_i^t + u_{i+n}^t\}\{v_{i+1}^t - v_i^t\}$$
$$+ \{u_{i-1}^t + u_{i+n-1}^t\}\{v_i^t - v_{i-1}^t\}]$$
$$= \frac{g}{R\Delta\varphi}\{\zeta_i^{t+\Delta t} - \zeta_{i+n}^{t+\Delta t}\}$$
$$- \frac{k}{e_i^t}v_i^{t+\Delta t}[\{\tilde{u}_i^t\}^2 + \{v_i^t\}^2]^{1/2} + \frac{1}{\rho}\left\{-Q_i^t + \frac{G_i^t}{e_i^t}\right\}$$
$$+ \frac{A_h}{R^2\cos^2\varphi_{m+1}\Delta\chi^2}[\{v_{i+1}^t - 2v_i^t + v_{i-1}^t\}]$$
$$+ \frac{A_h}{R^2\cos\varphi_{m+1}\Delta\varphi^2}[\cos\varphi_m\{v_{i-n}^t - v_i^t\}$$
$$+ \cos\varphi_{m+2}\{v_i^t - v_{i+n}^t\}], \quad \text{(A3)}$$

where

$$\tilde{u}_i^t = 0.25\{u_i^t + u_{i-1}^t + u_{i+n}^t + u_{i+n-1}^t\}$$

u at v point i and time t

$$\tilde{v}_i^t = 0.25\{v_i^t + v_{i-1}^t + v_{i+n}^t + v_{i+n-1}^t\}$$

v at u point i and time t

$$P_i^t = \frac{1}{R\cos\varphi}\frac{\partial p_a}{\partial\chi} \text{ at } u \text{ point } i \text{ and time } t$$

$$Q_i^t = \frac{1}{R}\frac{\partial p_a}{\partial\varphi} \text{ at } v \text{ point } i \text{ and time } t$$

$$F_i^t = F_s \text{ at } u \text{ point } i \text{ and time } t$$

$$G_i^t = G_s \text{ at } v \text{ point } i \text{ and time } t.$$

The terms

$$\frac{uv\tan\varphi}{R}$$

and

$$\frac{u^2\tan\varphi}{R},$$

in (1) and (2), respectively, are ignored in the present implementation of the model.

The drying scheme of Flather and Heaps (1975) is used with (A2) and (A3). Thus, at any u point,

if (i) $D_i^{t+\Delta t} > 0$ and $D_{i+1}^{t+\Delta t} > 0$, so that $d_i^{t+\Delta t} > 0$

or (ii) $D_i^{t+\Delta t} > 0$ and $D_{i+1}^{t+\Delta t} \leqslant 0$, and $d_i^{t+\Delta t} > 0$ and $\zeta_i^{t+\Delta t} - \zeta_{i+1}^{t+\Delta t} > \epsilon$

or (iii) $D_i^{t+\Delta t} \leqslant 0$ and $D_{i+1}^{t+\Delta t} > 0$, and $d_i^{t+\Delta t} > 0$ and $\zeta_{i+1}^{t+\Delta t} - \zeta_i^{t+\Delta t} > \epsilon$,

then (A2) is solved for $u_i^{t+\Delta t}$; otherwise set $u_i^{t+\Delta t} = 0$. Similarly, at any v point,

if (i) $D_i^{t+\Delta t} > 0$ and $D_{i+n}^{t+\Delta t} > 0$, so that $e_i^{t+\Delta t} > 0$

or (ii) $D_i^{t+\Delta t} > 0$ and $D_{i+n}^{t+\Delta t} \leqslant 0$, and $e_i^{t+\Delta t} > 0$ and $\zeta_i^{t+\Delta t} - \zeta_{i+n}^{t+\Delta t} > \epsilon$

or (iii) $D_i^{t+\Delta t} \leqslant 0$ and $D_{i+n}^{t+\Delta t} > 0$, and $e_i^{t+\Delta t} > 0$ and $\zeta_{i+n}^{t+\Delta t} - \zeta_i^{t+\Delta t} > \epsilon$,

then (A3) is solved for $v_i^{t+\Delta t}$; otherwise set $v_i^{t+\Delta t} = 0$. This allows inundation of coastal areas or intertidal banks, as in the original scheme of Flather and Heaps (1975), and, in combination with the modified continuity equation (1) and (A1), allows 1D flow along a channel to transform into 2D flow when the bank level is overtopped.

On open sea boundaries, which in the present scheme follow the sides of grid elements (as shown in Fig. A3), the "radiation" condition (9) replaces the appropriate equation of motion, (A2) or (A3), and is used to compute the normal component of current across the boundary. Thus, for a u point located on an *eastern* open boundary, (9) is approximated by

$$u_i^{t+\Delta t} = \hat{u}_i^{t+\Delta t} + \left(\frac{g}{d_i^{t+\Delta t}}\right)^{1/2}\{\zeta_i^{t+\Delta t} - \hat{\zeta}_i^{t+\Delta t}\}. \quad \text{(A4)}$$

The equivalent formula for a u point on a *western* boundary is

$$u_i^{t+\Delta t} = \hat{u}_i^{t+\Delta t} - \left(\frac{g}{d_i^{t+\Delta t}}\right)^{1/2}\{\zeta_{i+1}^{t+\Delta t} - \hat{\zeta}_{i+1}^{t+\Delta t}\}. \quad \text{(A5)}$$

In (A4) and (A5), the current across the open boundary is determined from the elevation at the first ζ point within the model. Note the different signs in (A4) and (A5), which are such that an excess of *computed* elevation, $\zeta_i^{t+\Delta t}$, over the *specified* input value, $\hat{\zeta}_i^{t+\Delta t}$, generates additional flow *out* of the model. The assumption is that the difference between the computed and specified disturbance at an open boundary is generated within the model and propagates out across the boundary as a progressive wave. Equivalent equations apply to v points on northern and southern boundary segments. In addition, since not all discrete values of the variables required in the advective terms at u and v points are within one-half grid distance of open boundaries, these terms are omitted from (A2) and (A3) at such locations.

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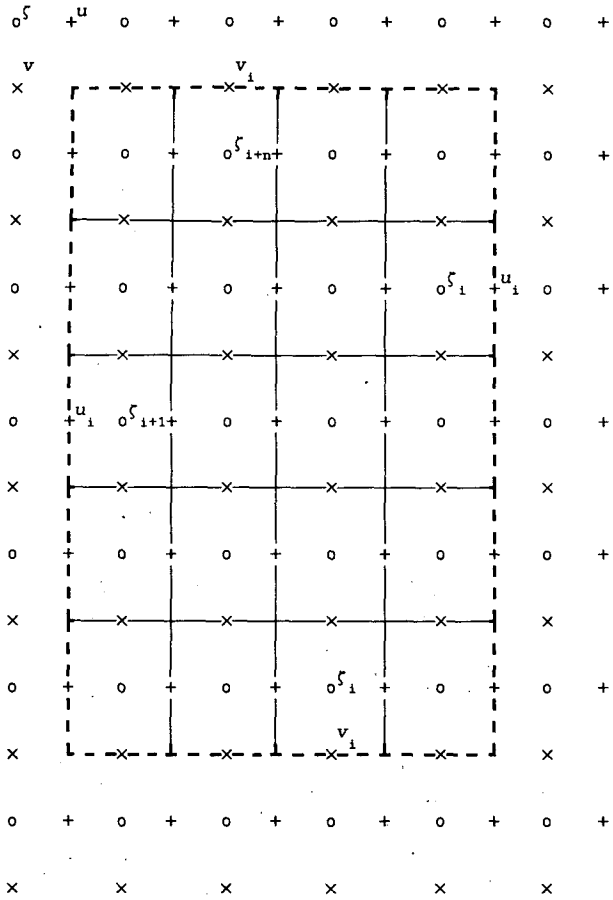

FIG. A3. Open boundaries (heavy broken lines) follow grid box sides.

The order of calculations in a complete cycle, starting from initial values of ζ, u, and v defined at every point of the grid, is as follows:

1. Solve (A1) at all interior ζ points to derive $\zeta_i^{t+\Delta t}$ from ζ_i^t.
2. Update all open boundary currents by solving (A4), (A5), and equivalent equations with prescribed input values (including tide and surge components).
3. Solve (A2) at all interior u points to derive $u_i^{t+\Delta t}$ from u_i^t.
4. Solve (A3) at all interior v points to derive $v_i^{t+\Delta t}$ from v_i^t.

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